1) $$z = \frac{a+bj}{c+dj}\left(\frac{c-dj}{c-dj}\right)$$

$$= \frac{(a+bj)(c-dj)}{c^2+d^2}$$

$$= \frac{ac-adj+bcj+bd}{c^2+d^2}$$

$$= \frac{1}{c^2+d^2}\left[ac+bd+(bc-ad)j\right]$$

Cartesian form $= \dfrac{ac+bd}{c^2+d^2} + \dfrac{bc-ad}{c^2+d^2}j$

Polar form

$$= \frac{1}{c^2+d^2}\sqrt{(ac+bd)^2+(bc-ad)}\,e^{j\arctan\left(\frac{bc-ad}{ac+bd}\right)}$$

1) $$z = \frac{1}{c+dj}\left(\frac{c-dj}{c-dj}\right)$$

$$= \frac{c-dj}{c^2+d^2}$$

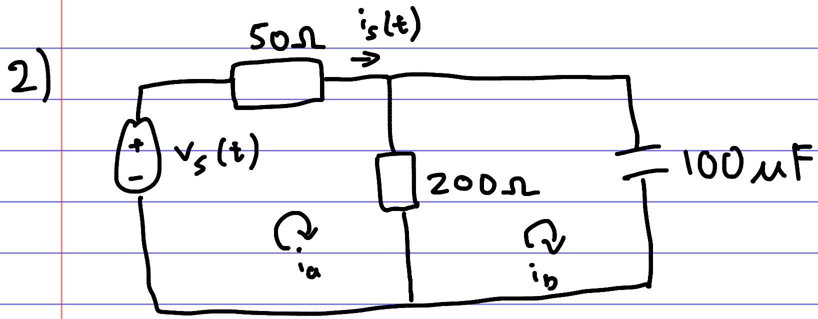
$$= \frac{1}{c^2+d^2}\left(c-dj\right)$$

Cartesian form $= \frac{c}{c^2+d^2} - \frac{d}{c^2+d^2}j$

Polar form $= \frac{1}{c^2+d^2}\sqrt{c^2+d^2}\, e^{j\arctan\left(\frac{-d}{c}\right)}$

$$= \frac{1}{\sqrt{c^2+d^2}}\, e^{-j\arctan\left(\frac{d}{c}\right)}$$

2)



$Z_{50\Omega} = 50\ \Omega$

$v_s(t) = 10\cos(100t)$

$= 10 \angle 0°$

$Z_{200\Omega} = 200\ \Omega$

$$Z_{100\mu F} = \frac{1}{j\omega C}$$

$$= \frac{1}{j(100)(100\times10^{-6})}$$

$$= \frac{100}{j}$$

$$= -100j\ \Omega$$

$$Z_{equiv} = \left(\frac{1}{Z_{200\Omega}} + \frac{1}{Z_{100\mu F}}\right)^{-1}$$

$$= \left(\frac{1}{200} + \frac{1}{-100j}\right)^{-1}$$

$$= 40 - 80j\ \Omega$$

2) $i_s = \dfrac{v_s}{Z_T}$

$= \dfrac{10}{50+40-80j}$

$= 0.0830454798 \angle 41.63353935^\circ$

$\approx 0.083 \angle 41.6^\circ$

$\approx 0.083 \cos(100t + 41.6^\circ)$